Machine Learning

15-05-26

XGBoost

- In XGB, trees are built in series but nodes are split in parallel → making training faster
- The nature of DTrees built in GBM vs XGB is different
- In XGB model regularization is inbuilt which avoids overfitting
- GBM doesn't handle null/missing values internally
- XGB handles null values internally. No need of handling null values in preprocessing steps

XGBoost Regression:

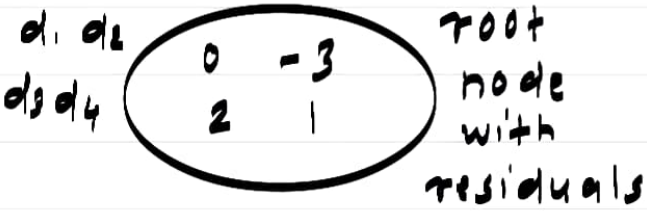
	x_1	x_2	y	$\hat{y}_0$	$R_1 = y - \hat{y}_0$
d_1			1	1	1-1 = 0
d_2			-2	1	-2-1 = -3
d_3			3	1	3-1 = 2
d_4			2	1	2-1 = 1

$\bar{y} = \frac{4}{4} = 1$

$\hat{y}_0 = \bar{y}$

train extreme DTree model m1 with x_1, x_2 as features and R_1 as target

- In extreme DTree, similarity gain is used for identifying the best decision rule. (instead of information gain)



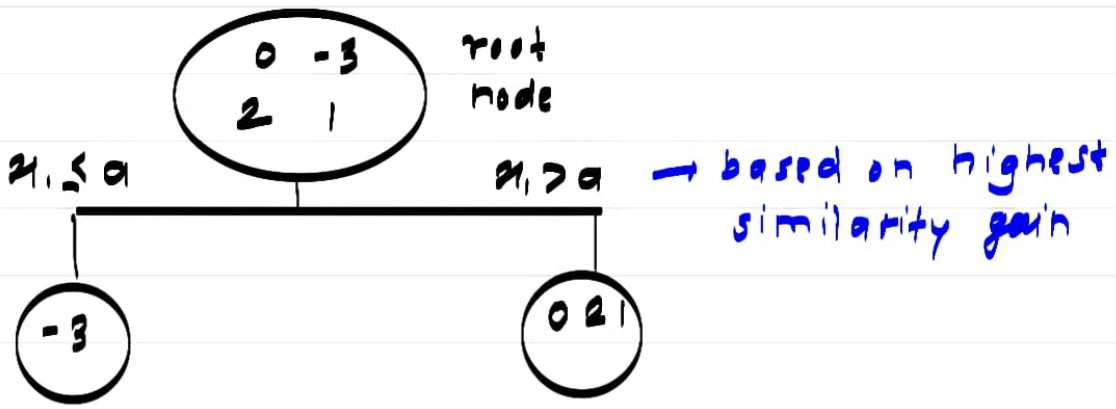
$$\text{similarity score for a node} = \frac{[\Sigma \text{Residuals}]^2}{\text{no. of residuals in the node} + \lambda}$$

$\lambda \rightarrow$ inbuilt regularization parameter in extreme DTree

$\lambda \uparrow \rightarrow$ regularization $\uparrow \rightarrow$ overfiting $\downarrow$

$$\text{similarity gain} = S_{child1} + S_{child2} - S_{parent}$$

- the decision rule giving highest similarity gain is used to split the parent node.



$\lambda = 1$

$$S_{parent} = \frac{[0 + (-3) + 2 + 1]^2}{4 + 1} = \frac{0}{5} = 0$$

$$S_{child1} = \frac{[-3]^2}{1+1} = \frac{9}{2} = 4.5$$

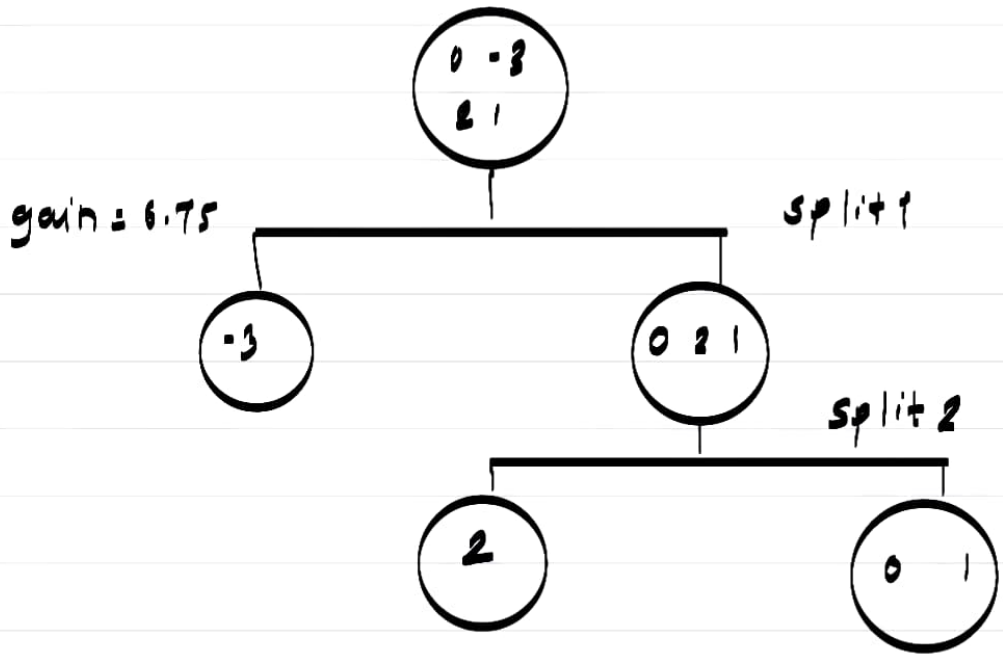
$$S_{child2} = \frac{[0+2+1]^2}{3+1} = \frac{9}{4} = 2.25$$

similarity gain : $S_{child1} + S_{child2} - S_{parent}$

$= 4.5 + 2.25 - 0$

$= 6.75$

one more hyperparameter γ gamma is set in XGB which avoids sub-optimal splits to reduce overfiting



gain for split 2 : $S_{child1} + S_{child2} - S_{parent}$

$= \frac{[2]^2}{1+1} + \frac{[0+1]^2}{2+1} - \frac{[0+2+1]^2}{3+1}$

$= 4/2 + 1/3 - 9/4$

$= 2 + 0.33 - 2.25$

$= 0.08$

if we set $\gamma = 2$ then

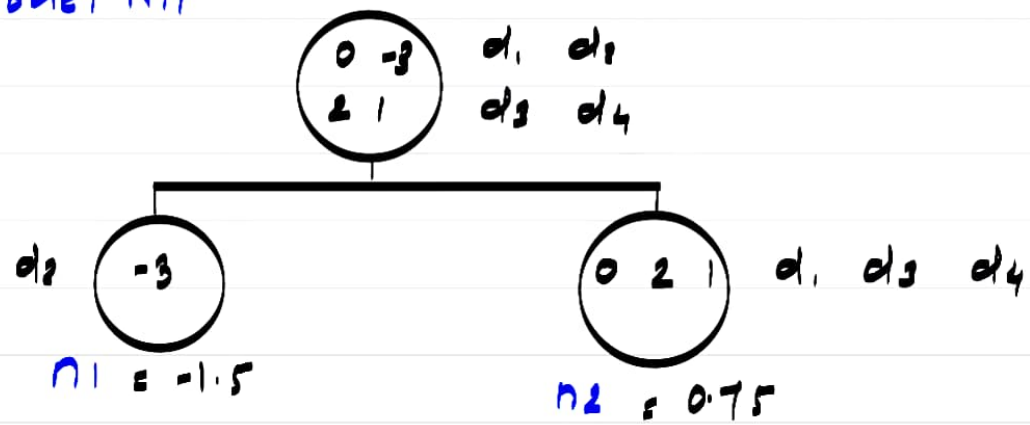
gain = 0.8 $< \gamma$ → makes split2 sub-optimal

hence
split2 will be pruned (no split will happen for that parent)

sub-optimal splits cause overfiting and
γ prevents it
this is called prunning in XGB
γ is prunning parameter

higher γ value ⇒ higher prunning rate

final model m1



$$\text{output from a node} = \frac{\Sigma \text{Residuals}}{\text{no. of residuals in node} + \lambda}$$

$$\text{output from node } n1 = \frac{-3}{1+1} = -1.5$$

$$\text{output from node } n2 = \frac{0+2+1}{3+1} = \frac{3}{4} = 0.75$$

$\alpha = 0.5$

	x_1	x_2	y	$\hat{y}_0$	$R_1 = y - \hat{y}_0$	$\hat{R}_1$	$\hat{y}_1 = \hat{y}_0 + \alpha \hat{R}_1$
d_1		1	1	$1-1=0$	0.75	$1 + 0.5 \times 0.75$	$= 1.375$
d_2		-2	1	$-2-1=-3$	-1.5	$1 + 0.5 \times (-1.5)$	$= 0.25$
d_3		3	1	$3-1=2$	0.75	$1 + 0.5 \times 0.75$	$= 1.375$
d_4		2	1	$2-1=1$	0.75	$1 + 0.5 \times 0.75$	$= 1.375$

compute $R_2 = y - \hat{y}_1$

	R_2
d_1	$1 - 1.375 = -0.375$
d_2	$-2 - 0.25 = -2.25$
d_3	$3 - 1.375 = 1.625$
d_4	$2 - 1.375 = 0.625$

train next XDTree model m_2 with
x_1, x_2 as features and R_2 as target



lets say we trained 4 models in this series

new datapoint



final prediction from ensemble $= \hat{y}_4 = \hat{y}_3 + \alpha \hat{R}_4$

$= \hat{y}_2 + \alpha \hat{R}_3 + \alpha \hat{R}_4$

$= \hat{y}_1 + \alpha \hat{R}_2 + \alpha \hat{R}_3 + \alpha \hat{R}_4$

$= \hat{y}_0 + \alpha [\hat{R}_1 + \hat{R}_2 + \hat{R}_3 + \hat{R}_4]$

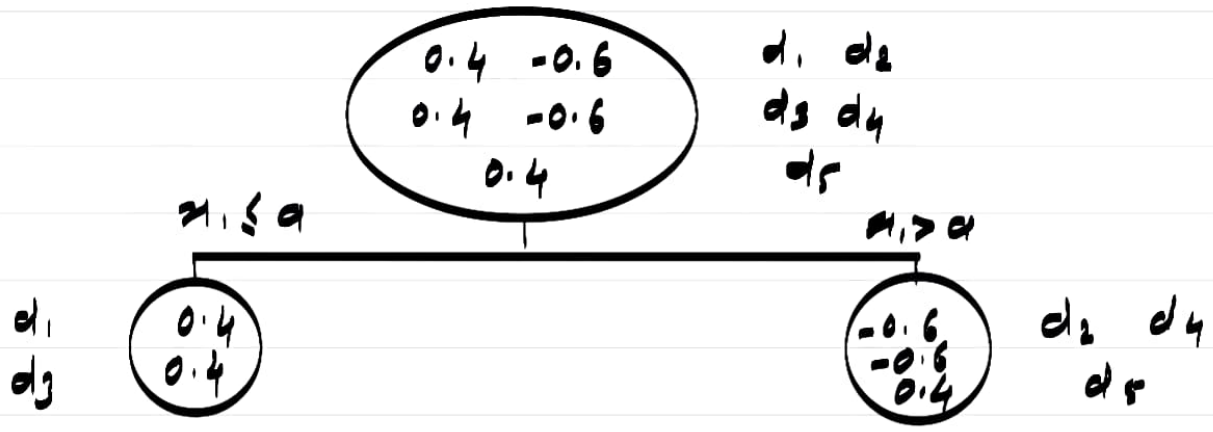
XGBoost classifier:

	x_1	x_2	y	y	$Z_0 = \ln(\text{odds})$	$\hat{y}_0 = \sigma(Z_0)$	$R_1 = y - \hat{y}_0$
d_1			C_1	1	0.4	$1/1+e^{-0.4} = 0.6$	$1 - 0.6 = 0.4$
d_2			C_2	0	0.4	0.6	$0 - 0.6 = -0.6$
d_3			C_1	1	0.4	0.6	0.4
d_4			C_2	0	0.4	0.6	-0.6
d_5			C_1	1	0.4	0.6	0.4

$$\ln(\text{odds}) = \ln\left[\frac{p(+ve)}{p(-ve)}\right] = \ln(3/2) = 0.4$$

train XDTree model m_1, with x_1, x_2 as features and R_1 as target

XDTree model will be trained with splits giving highest similarity gain



$$\text{similarity score of a node} = \frac{[\Sigma \text{Residuals}]^2}{\{\Sigma \text{prev.prob}\,(1-\text{prev prob})\} + \lambda}$$

lets put $\lambda = 2$

$$S_{parent} = \frac{[0.4 + (-0.6) + 0.4 + (-0.6) + 0.4]^2}{0.6 \times (1-0.6) + 0.6\,(1-0.6) + 0.6\,(1-0.6) + 0.6\,(1-0.6) + 0.6\,(1-0.6)} + 2$$

$$= 0$$

$$S_{child1} = \frac{[0.4 + 0.4]^2}{0.6\,(1-0.6) + 0.6\,(1-0.6) + 2} = 0.25$$

$$S_{child2} = \frac{[0.4 + (-0.6) + (-0.6)]^2}{0.6 \times (1-0.6) + 0.6\,(1-0.6) + 0.6\,(1-0.6) + 2} = 0.23$$

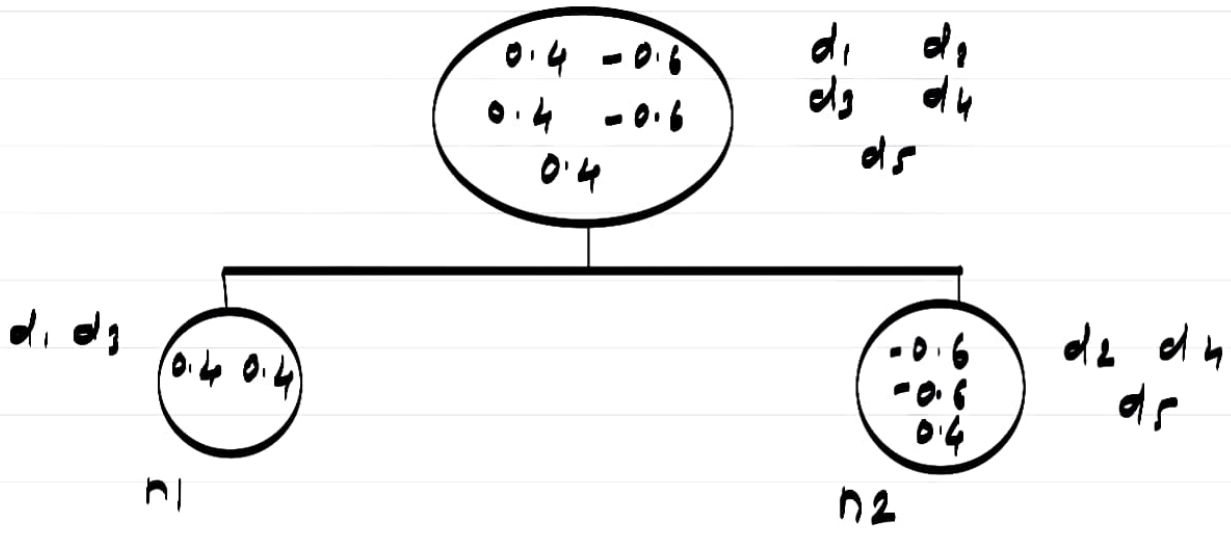
$$\text{similarity gain} = S_{child1} + S_{child2} - S_{parent}$$

$$= 0.25 + 0.23 - 0$$

$$= 0.48$$

split with highest similarity gain is the best split

regularization using γ gamma is also applied to avoid overfiting



output of the node = $\frac{\sum \text{Residuals}}{\sum \text{prev.prob}(1-\text{prev.prob}) + \lambda}$

output of n1 = $\frac{0.4+0.4}{0.6(1-0.6)+0.6(1-0.6)+2} = 0.32$

output of n2 = $\frac{0.4+(-0.6)+(-0.6)}{0.6(1-0.6)+0.6(1-0.6)+2+0.6(1-0.6)} = -0.3$

$\alpha = 1$

	x_1	x_2	y	Z_0	$\hat{y}_0$	R_1	$\hat{R}_1$	$Z_1 = Z_0 + \alpha \hat{R}_1$	$\hat{y}_1 = \sigma(Z_1)$
d1			1	0.4	0.6	0.4	0.32	$0.4+1\times0.32 = 0.72$	0.67
d2			0	0.4	0.6	-0.6	-0.3	$0.4+1\times-0.3 = 0.1$	0.52
d3			1	0.4	0.6	0.4	0.32	$0.4+1(0.32) = 0.72$	0.67
d4			0	0.4	0.6	-0.6	-0.3	$0.4+1(-0.3) = 0.1$	0.52
d5			1	0.4	0.6	0.4	-0.3	$0.4+1(-0.3) = 0.1$	0.52

α - learning rate

$R_2 = y - \hat{y}_1$

d_1	$1 - 0.67 = 0.33$
d_2	$1 - 0.52 = 0.48$
d_3	0.33
d_4	0.48
d_5	0.48

train model m_2 with x_1, x_2 as features and R_2 as target

$R_2 \rightarrow [m_2]$

$\downarrow$

$\hat{R}_2 \rightarrow Z_2 = Z_1 + \alpha \hat{R}_2$

$\downarrow$

$\hat{y}_2 = \sigma(Z_2)$

$\downarrow$

$R_3 = y - \hat{y}_2$

$\rightarrow [m_3]$

$\downarrow$

$\hat{R}_3 \rightarrow Z_3 = Z_2 + \alpha \hat{R}_3$

$\downarrow$

$\hat{y}_3 = \sigma(Z_3)$

$\vdots$

lets say we built 4 models in series

final XGB prediction = $\hat{y}_4 = \sigma(Z_4)$

$= \sigma(Z_3 + \alpha \hat{R}_4)$

if $\hat{y}_4 = 0.37$ and threshold: 0.6 then

as $\hat{y}_4 <$ threshold, predicted class : -ve = C_2